

CS 6120 Assignment 2

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Q2.1 — Loss function gradient $\partial L / \partial \hat{y}$

The binary cross-entropy loss is

$$L(\hat{y}, y) = -\frac{1}{N} \sum_{i=1}^N \left[y^{(i)} \log \hat{y}^{(i)} + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)}) \right].$$

Differentiating with respect to a single prediction $\hat{y}^{(i)}$ (only the i -th term in the sum depends on it):

$$\frac{\partial L}{\partial \hat{y}^{(i)}} = -\frac{1}{N} \frac{\partial}{\partial \hat{y}^{(i)}} \left[y^{(i)} \log \hat{y}^{(i)} + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)}) \right].$$

Applying $\frac{d}{du} \log(u) = \frac{1}{u}$ and the chain rule on $\log(1 - \hat{y}^{(i)})$:

$$\frac{\partial}{\partial \hat{y}^{(i)}} [y^{(i)} \log \hat{y}^{(i)}] = \frac{y^{(i)}}{\hat{y}^{(i)}}, \quad \frac{\partial}{\partial \hat{y}^{(i)}} [(1 - y^{(i)}) \log(1 - \hat{y}^{(i)})] = -\frac{1 - y^{(i)}}{1 - \hat{y}^{(i)}}.$$

Combining:

$$\boxed{\frac{\partial L}{\partial \hat{y}^{(i)}} = -\frac{1}{N} \left(\frac{y^{(i)}}{\hat{y}^{(i)}} - \frac{1 - y^{(i)}}{1 - \hat{y}^{(i)}} \right)}$$

Q2.2 — Sigmoid gradient

Starting from the definition

$$\sigma(z) = \frac{1}{1 + e^{-z}} = (1 + e^{-z})^{-1}.$$

Apply the chain rule, using $\frac{d}{dz} e^{-z} = -e^{-z}$:

$$\frac{\partial \sigma(z)}{\partial z} = -(1 + e^{-z})^{-2} \cdot (-e^{-z}) = \frac{e^{-z}}{(1 + e^{-z})^2}.$$

Rewrite $e^{-z} = (1 + e^{-z}) - 1$ in the numerator and split the fraction:

$$\frac{e^{-z}}{(1 + e^{-z})^2} = \frac{1 + e^{-z}}{(1 + e^{-z})^2} - \frac{1}{(1 + e^{-z})^2} = \frac{1}{1 + e^{-z}} - \left(\frac{1}{1 + e^{-z}} \right)^2 = \sigma(z) - \sigma(z)^2.$$

Factor out $\sigma(z)$:

$$\boxed{\frac{\partial \sigma(z)}{\partial z} = \sigma(z)(1 - \sigma(z))}$$

Q2.3 — Gradients for a single layer $\hat{y} = \sigma(Wh + b)$

Here h denotes the *input* to this layer. For the first layer of the network $h = x$ (the raw input); for any subsequent layer, h is the activation produced by the previous layer. The derivation below is written generically so it applies in either case.

Let $z = Wh + b$, so that $\hat{y} = \sigma(z)$. By Q2.2,

$$\frac{\partial \hat{y}}{\partial z} = \sigma(z)(1 - \sigma(z)) = \hat{y}(1 - \hat{y}).$$

Gradient with respect to W . Since W_{ij} appears as $W_{ij}h_j$ in $z_i = \sum_k W_{ik}h_k + b_i$ (and b_i does not depend on W_{ij}), its derivative is h_j :

$$\frac{\partial z_i}{\partial W_{ij}} = h_j.$$

Applying the chain rule (transposing h since (output dim,) $\cdot$ (input dim,) do not form a matrix):

$$\boxed{\frac{\partial \hat{y}}{\partial W} = \hat{y}(1 - \hat{y}) h^\top}$$

Gradient with respect to h . From $z = Wh + b$ we have $\frac{\partial z}{\partial h} = W$ (transposing W since (output dim, input dim) $\cdot$ (output dim,) does not align):

$$\boxed{\frac{\partial \hat{y}}{\partial h} = W^\top \hat{y}(1 - \hat{y})}$$

Gradient with respect to b . Since b adds directly to z (entry by entry), nudging any b_i changes z_i by exactly the same amount, so the gradient passes through unchanged:

$$\boxed{\frac{\partial \hat{y}}{\partial b} = \hat{y}(1 - \hat{y})}$$

Q2.4 — All parameter gradients

The forward computation of the two-layer network is

$$z_1 = W_1x + b_1, \quad h = \sigma(z_1), \quad z_2 = W_2h + b_2, \quad \hat{y} = \sigma(z_2).$$

Working backwards from the loss, applying the chain rule at each step.

Step 1 — Gradient at z_2 . Combining Q2.1 (loss gradient) with Q2.2 (sigmoid gradient):

$$\frac{\partial L}{\partial z_2} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_2} = -\frac{1}{N} \left(\frac{y}{\hat{y}} - \frac{1-y}{1-\hat{y}} \right) \cdot \hat{y}(1 - \hat{y}).$$

Distributing $\hat{y}(1 - \hat{y})$ and simplifying (the fractions cancel):

$$\boxed{\frac{\partial L}{\partial z_2} = \frac{1}{N}(\hat{y} - y)}$$

Step 2 — Gradients for the second layer parameters. Using the single-layer rules from Q2.3 with input h :

$$\boxed{\frac{\partial L}{\partial W_2} = \frac{\partial L}{\partial z_2} h^\top} \quad \boxed{\frac{\partial L}{\partial b_2} = \sum_i \frac{\partial L}{\partial z_2^{(i)}}$$

The sum for b_2 appears because b_2 is used once per example in the batch, so its gradient collects one piece from each example and adds them up.

Step 3 — Push the gradient back to z_1 . Through W_2 and the first sigmoid:

$$\frac{\partial L}{\partial h} = W_2^\top \frac{\partial L}{\partial z_2}, \quad \frac{\partial L}{\partial z_1} = \frac{\partial L}{\partial h} \odot h(1-h),$$

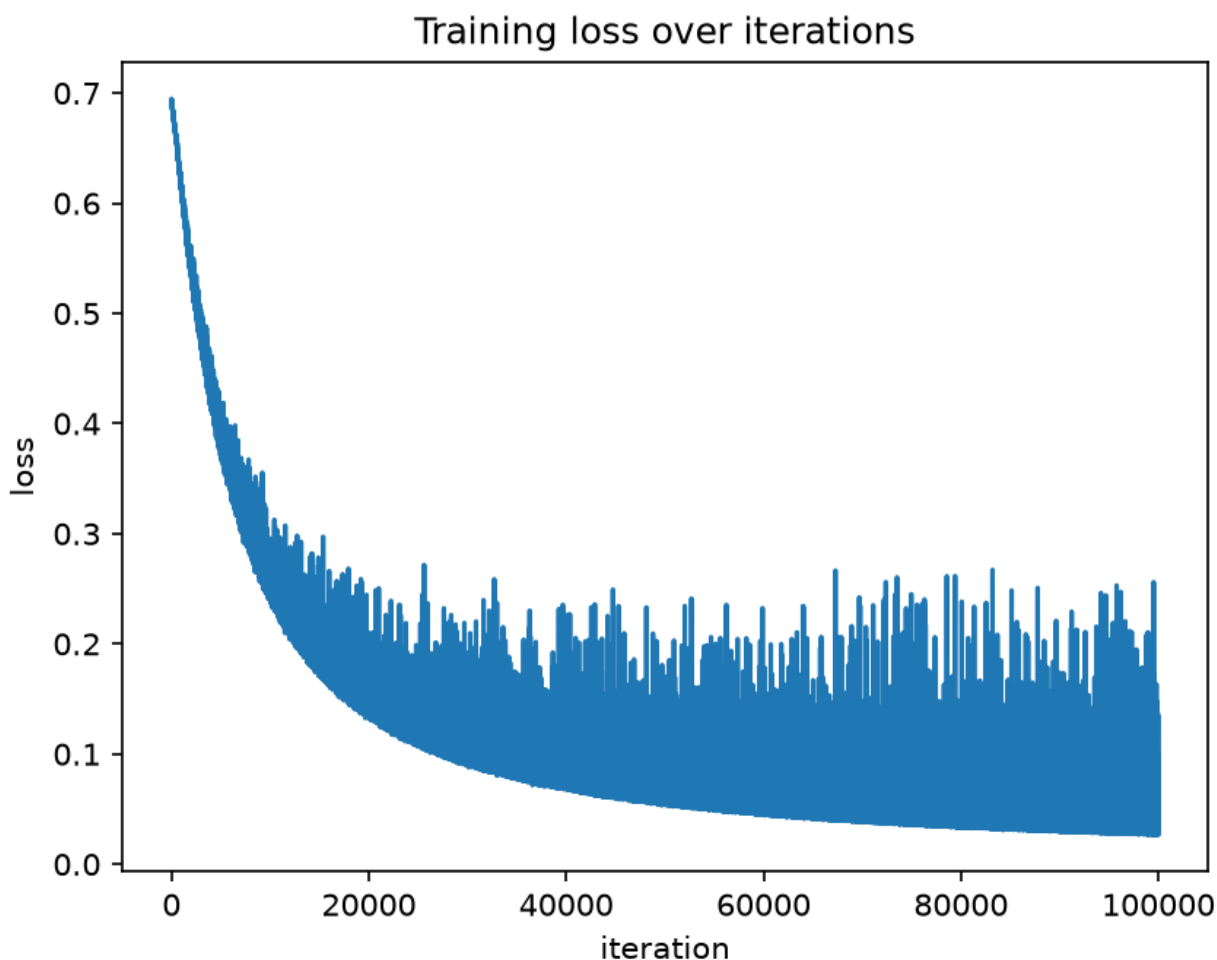
where $\odot$ means entry-by-entry multiplication, since each entry of z_1 passes through its own sigmoid independently.

Step 4 — Gradients for the first layer parameters. Applying Q2.3 with input x :

$$\boxed{\frac{\partial L}{\partial W_1} = \frac{\partial L}{\partial z_1} x^\top} \quad \boxed{\frac{\partial L}{\partial b_1} = \sum_i \frac{\partial L}{\partial z_1^{(i)}}$$

Same reasoning as for b_2 : b_1 is used once per example in the batch, so its gradient sums the contributions from all examples.

Q5 — Training results



Test accuracy

Train: 0.9961

Test: 0.9970

Sample predictions

Tweet: I had an amazing day! Everything went perfectly.
 $\hat{y}$: 0.9384 | Sentiment: Positive

Tweet: This dish is pushing the boundaries of what counts as food!
 $\hat{y}$: 0.2845 | Sentiment: Negative

Tweet: The weather is nice today :)
 $\hat{y}$: 0.9698 | Sentiment: Positive

Tweet: I'm so disappointed with the service, worst experience ever.
 $\hat{y}$: 0.1458 | Sentiment: Negative

Tweet: This is a wicked movie. The plot was terrible and I was sad until the ending!
 $\hat{y}$: 0.0248 | Sentiment: Negative